SWAFL Desktop 1.1.0: An Offline, Equation-Traceable Decision-Support Suite from Field Sketch and Satellite Indices to Irrigation Design, Daisy Simulation and Deterministic Phenotyping

Naziru Halilu[1,2,*]

[1] Agricultural University of Athens, Athens, Greece

[2] Public University of Navarre, Pamplona, Spain

ORCID: 0009-0006-2324-9599

[*] Correspondence: halilunaziru73@gmail.com

Abstract

The tools that a person needs to plan irrigation on one field are usually separate programs: a map, a source of satellite data, a crop-water calculator, a hydraulic design sheet, a simulation model, each with its own inputs and none of them showing where its numbers come from. This article describes SWAFL Desktop 1.1.0, an open, offline-first suite that joins these steps in one application and keeps one rule throughout: every number leads to a numbered equation with a provenance label (literature, data-driven or engineered). The suite has 22 decision layers in nine groups (irrigation design for surface, drip and sprinkler systems; irrigation control; mulch and evaporation; soil structure; deterministic phenotyping; field water balance; seasonal data; crops; feedback loops), two further layers that show Daisy results, a library of 48 crops with 110 crop-and-density combinations, and tools for the real field: a sketch of the field and the irrigation system, the satellite indices NDVI, EVI, SAVI, NDRE and NDWI from Sentinel-2, image analysis of the satellite photograph, water sources, terrain and soil from open data, management zones, zone-by-zone variable-rate irrigation, a field report and prescription files. The computing engines are the meta-model of a 10 190-run Daisy ensemble, a daily 40-layer water balance and closed-form hydraulic modules; a graphical workbench runs Daisy itself (Papers 4 and 5). A virtual phenotyping experiment grows one plant of a chosen crop and genotype in the soil, weather and management of the project, deterministically, well-watered and under drought. We report what was tested for this release. All 22 layers open and run in a Chromium browser without a script error, and each produces a dashboard with results, charts, a design or figure tab and messages. The 48-crop library is internally consistent except that 13 of its 110 evergreen combinations have a mid-season coefficient below the initial one, which follows the source values. The

phenotyping model is bit-for-bit deterministic over 28 runs and closes its soil water balance to 4×10^{-5} mm; under a drought pulse seven genotypes lose 14 to 42 % of the grain mass and water-saving, deep-rooting and stay-green genotypes lose least. The manual has about 470 numbered equations. No field validation is claimed; the satellite and open-data services could not be reached in the environment of the tests, so these paths were exercised on generated test data only.

Keywords. decision support; irrigation design and scheduling; Sentinel-2 vegetation indices; field sketch; plant phenotyping; crop water requirement; offline software; equation traceability; open science

1. Introduction

Irrigation planning on a farm combines at least five kinds of work. The field has to be found, outlined and described (its area, slope, soil, water source). Its crop has to be monitored, today by satellite. The irrigation system has to be designed and checked hydraulically. The schedule (when, how much) has to be calculated from the weather, the crop and the soil. And the decision has to be put against what a process model says would happen if the soil, the weather or the management were different. Each kind of work has good tools: web maps and the Sentinel-2 archive (Drusch et al., 2012), crop-water calculators built on the FAO procedures (Allen et al., 1998; Doorenbos and Kassam, 1979), hydraulic design methods (Keller and Bliesner, 1990; Walker, 1989), simulation models for crops and soil water (Hansen et al., 2012; Jones et al., 2003; Keating et al., 2003; Šimůnek et al., 2008; Steduto et al., 2009). They rarely share inputs, they often need a connection to the internet or a licence, and in many of them the path from an input to an output cannot be followed.

This article describes one application that joins them and the tests that were made of it for release 1.1.0. Its design goals are four. *Offline first*: the core computes without a connection; open data (satellite scenes, maps, weather, soil, elevation) are used when a connection exists, and the program says what it used and what it could not reach. *Traceable*: every output leads to a numbered equation, and every equation carries a label of its origin, so that a reader can tell what is taken from the literature, what was fitted to data, and what was designed for the software. *One project*: the soil, crop, weather, mulch, irrigation and date are set once and every layer uses them. *Open*: the source, the manual, the data and the check scripts are public (Software, data and code availability).

The software paper of the first release (Paper 3) described the meta-model, the daily model and the closed-form design modules and verified them. Release 1.1.0 adds a graphical workbench for Daisy (Paper 4) and its connection to the decision layers (Paper 5), a deterministic phenotyping layer, a library of 48 crops, a sketch tool, tools for the real field with satellite indices, variable-rate irrigation and a field report. This article is the description of the whole and the report of what was tested.

Contributions. (1) The architecture of a suite in which the field, the design, the schedule, the process model and the phenotyping share one project and one rule of traceability. (2) The tools for the real field and the way they fall back when the data are not available. (3) A deterministic

phenotyping layer and its tests. (4) A crop library with a stated source of every coefficient. (5) A test of the whole suite (all layers opened and run; the crop library; the phenotyping model) and a statement of what was not tested.

What is new. The new element is the combination, under one rule of traceability: a field sketch, Sentinel-2 indices and image analysis, management zones with zone-by-zone irrigation, closed-form design of three irrigation systems, a 48-crop library with a stated source for every coefficient, a daily water balance, the Daisy workbench, and a deterministic virtual phenotyping layer that grows a plant in the soil, weather and management of the project, well-watered and under drought, without random numbers. The tests that go with it (all 22 layers run, crop constraints, repeatability and mass closure of the phenotyping model) are reported together with the limits of what they show.

2. The suite

2.1 Architecture

Figure 1 shows the architecture. The user works in one window with a tree of layers on the left, the picture or design of the layer in the middle, a dashboard on the right and a panel of project properties. The dashboard of every layer has the same five tabs (results, design, charts, figure, messages), so that a result is read in the same place whichever layer made it. A single Run button starts a layer; the results of a layer wait for a new run when a setting of the project is changed, so that nothing moves while the settings are being made. The date of the toolbar is the working date of the whole suite and can be set to any day from 1980 to 2100: the weather of 1980 to 1999 is a recorded baseline, that of 2000 to 2100 is projected from a chosen baseline year (a typical, dry, wet or specific year), and the user is told that projections are simulations and not forecasts.

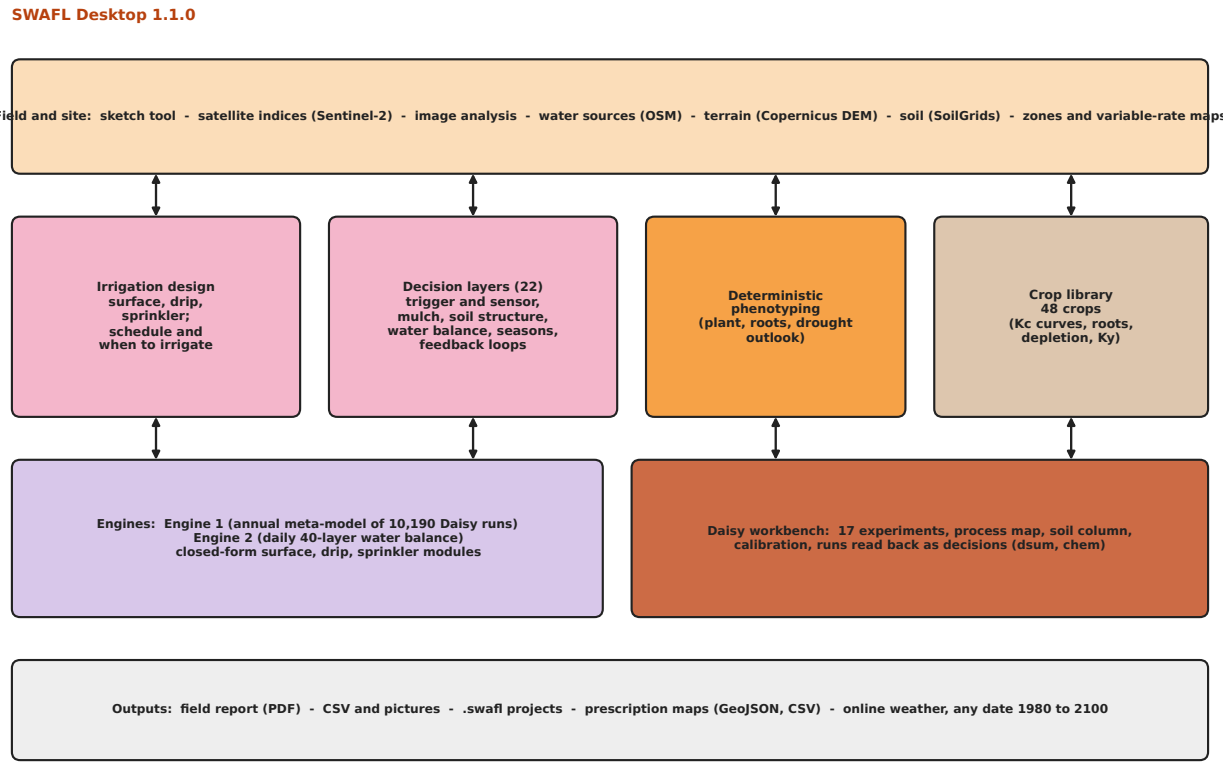


Figure 1: Architecture of SWAFL Desktop 1.1.0. The tools of the real field (top), the design and decision layers, the phenotyping layer and the crop library, the three computing engines and the Daisy workbench, and the outputs.

2.2 The layers

Table 1 lists the 22 layers. The first three are closed-form design modules for surface (border, furrow, basin), drip and sprinkler irrigation, with a layout drawing, a pipe network, pump and hydraulic grade line, design checks, an optimiser for the surface layout and a schedule that ends in a message on when to irrigate with the interval, the application efficiency and the storage efficiency. The layers of irrigation control, mulch and evaporation, and soil structure explore the findings of the Daisy ensemble (a depletion-based trigger law, the fate of the water that a mulch saves, a regime map of evaporation against drainage; Paper 1). The water-balance layers use the daily model; the seasonal layers run it over 20 seasons. Two more layers, the Daisy decision table and the fate of a chemical, show only the results of Daisy runs made in the workbench (Paper 5).

Table 1: The 22 decision layers of release 1.1.0 by group, and the two layers of Daisy output.

Group	Id	Layers
Irrigation design	3	surface (`sid`), drip (`drip`) and sprinkler (`spr`) design
Irrigation control	2	sensor and trigger (`v1`); trigger law (`n1`)
Mulch and evaporation	5	mulch against bare soil (`v4`); mulch control (`v6`); evaporation physics (`e1`); fate of saved water (`n2`); regime map (`n3`)
Soil structure and tillage	3	tillage and structure (`v3`); soil hydraulic drivers (`v7`); root structure (`r1`)
Deterministic phenotyping	1	phenotyping of a plant under well-watered and drought treatments (`dph`)
Field water balance	3	field balance (`field`); soil, plant and air (`spac`); profile evolution (`v2`)
Seasonal data	3	reliability over 20 seasons (`yby`); full water balance (`wb2`); nitrogen budget (`nb`)
Crops	1	crop water requirement (`crop`)
Feedback loops	1	seven soil–atmosphere feedback loops (`v5`)
Daisy output	2	decision table (`dsum`); fate of a chemical (`chem`)

2.3 Traceability

The manual of the release has about 470 numbered equations and 482 statements of provenance. Counted by label, 159 are marked as literature, 69 as data-driven and 287 as engineered (some equations carry two labels). A reader of a layer is told which equation produced a number, and which of the three origins it has. The label "engineered" means that the form of the relation was chosen for the software and has not been fitted to data or taken from a source; it is the largest class, and it is the class for which the verification of this article and of Paper 3 matters most.

3. The real field

3.1 Sketch

The field sketch is a drawing tool in which the user states the field (its corners on a grid or a rectangle by length and width, the ground slope, the direction of the laterals or furrows), places the water source, the inlet, the greenhouse, plant beds and areas that are not irrigated, and gets the area, the perimeter and the irrigated area. A five-step question list leads from the field to the crop, soil, water system and running; the sketch becomes the design of a layer with one click.

3.2 Satellite indices and image analysis

When the field is demarcated on the map (Esri World Imagery, OpenStreetMap), the suite searches the Sentinel-2 L2A catalogue (Element 84 Earth Search, with Microsoft Planetary Computer as the second source), reads the bands B02, B04, B05, B08 and B11 for the field only, removes the baseline offset and the pixels of cloud, shadow and snow with the scene classification, and computes for every 10 m cell

$$\begin{aligned} \mathrm{NDVI} &= \frac{\rho_N - \rho_R}{\rho_N + \rho_R}, \qquad \mathrm{EVI} = 2.5\,\frac{\rho_N - \rho_R}{\rho_N + 6\rho_R - 7.5\rho_B + 1}, \qquad \mathrm{SAVI} = \frac{(1+L)(\rho_N - \rho_R)}{\rho_N + \rho_R + L}, \\ \mathrm{NDRE} &= \frac{\rho_N - \rho_{RE}}{\rho_N + \rho_{RE}}, \quad \mathrm{NDWI} = \frac{\rho_N - \rho_{SW}}{\rho_N + \rho_{SW}} \qquad\qquad (L = 0.5), \end{aligned} \tag{1}$$

after Rouse et al. (1974), Huete et al. (2002), Huete (1988), Barnes et al. (2000) and Gao (1996). From the photograph of the field the visible-band indices VARI, GLI, ExG and NGRDI are averaged (Gitelson et al., 2002; Woebbecke et al., 1995), and a land cover (dense and sparse vegetation, bare soil, water, buildings and roads, shadow) is read by k-means clustering of six colour and texture features followed by named rules; this is classical computer vision and not a trained network, and the label of the equation says so. The field is split into five management zones by natural breaks of the smoothed index. A series of the scenes of the last twelve months (the least cloudy of each month, with a scene accepted when at least 30 % of the field is clear) gives for each cell a stability, a change since the previous scene and a trend, and groups the cells into three to six monitoring zones.

Open data complete the site: the water features around the field (OpenStreetMap) and their distance, the elevation and slope (Copernicus DEM), and the soil texture (SoilGrids; Poggio et al. 2021). The suite suggests where to take the water from and which irrigation system suits the field, with the reasons. If a service cannot be reached, the program says what it could not reach and uses the simulated indices of the suite, which mix the reflectance of crop and soil by the crop cover of the daily model.

3.3 Zone-by-zone irrigation, report and prescriptions

For each management zone the FAO-56 root-zone balance (Allen et al., 1998) is run with the zone's soil (SoilGrids at its centre, the nearest soil of the suite), its observed cover (the crop development of the daily model scaled by the zone's NDVI against the field's) and the weather

in use, which gives its own dates and depths of irrigation and the volume compared with that of one depth for the whole field. The result goes into a field report (PDF) and into prescription files (GeoJSON and CSV with the zone, date, net and gross depth of each cell). Further tools of the same panel estimate the ephemeral-gully and soil-loss risk from the terrain and the flow of water over it, and delineate watersheds and place soil sensors.

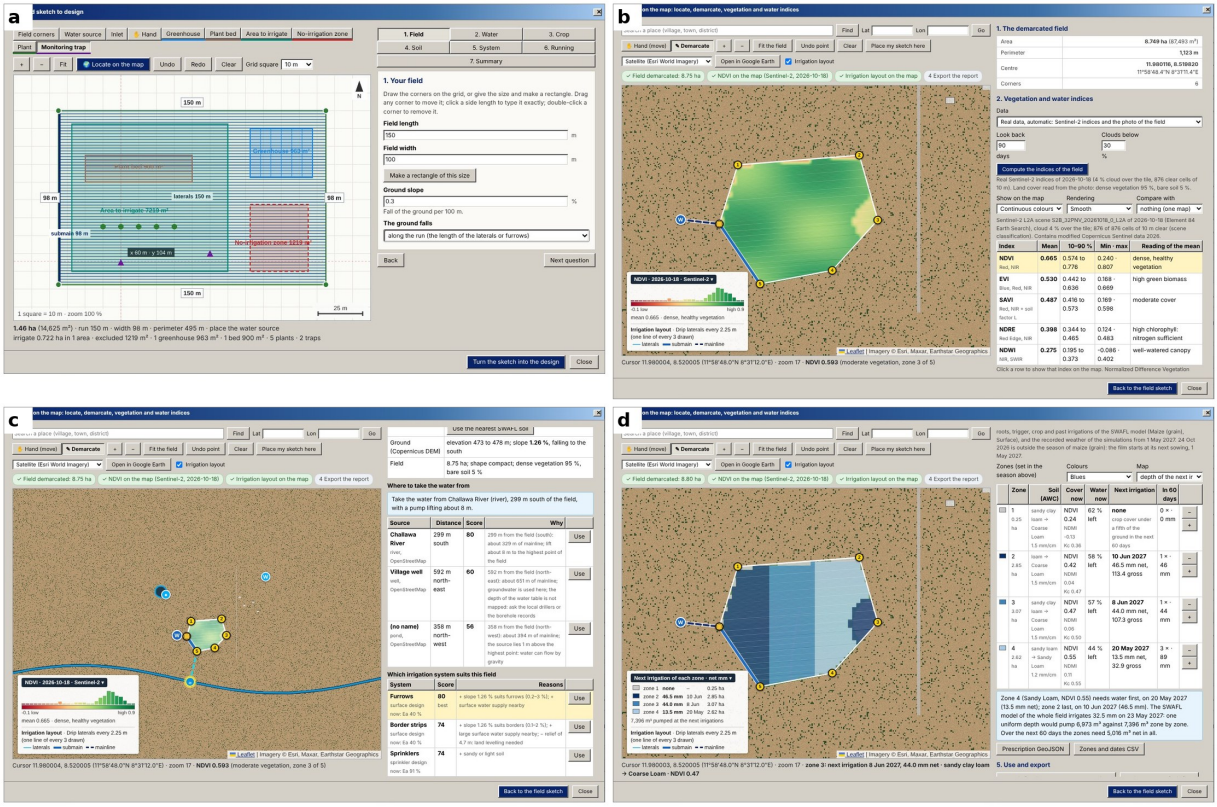


Figure 2: The tools of the real field (generated test image and test answers of the data services). (a) The field sketch with the inlet, the laterals, the greenhouse and an area that is not irrigated. (b) Sentinel-2 NDVI cut to the outline of the field with the table of indices. (c) Water sources and the suggested irrigation system. (d) The depth of the next irrigation of each management zone.

4. Crops

The library has 48 crops, 19 field crops (cereals, roots and tubers, legumes, oil crops, cotton) with a crop coefficient for the initial, mid-season and late-season stages, four stage lengths, height, rooting depth, the depletion fraction p and, where tabulated, the yield response factor K_y (Allen et al., 1998; Doorenbos and Kassam, 1979), and 29 perennial crops (tree, vine and berry crops) with several planting densities or training systems, each with its ground cover, height, mid-season

and end-season coefficient. The daily crop coefficient follows the piecewise curve of FAO-56,

$$K_c(t) = \begin{cases} K_{ini}, & t < L_1 \\ K_{ini} + (K_{mid} - K_{ini})\dfrac{t - L_1}{L_2}, & L_1 \le t < L_1 + L_2 \\ K_{mid}, & L_1 + L_2 \le t < L_1 + L_2 + L_3 \\ K_{mid} + (K_{end} - K_{mid}) \min\left(1, \dfrac{t - L_1 - L_2 - L_3}{L_4}\right), & \text{otherwise,} \end{cases} \tag{2}$$

which for evergreen crops is repeated each year. Every coefficient has a source field; where the root depth or the stages are not tabulated for a crop the group default is used and marked (11 crops); the yield response factor is tabulated for 11 crops and not tabulated for 37.

5. Deterministic phenotyping

The phenotyping layer is a virtual experiment. One plant of a chosen crop (modelled as maize, sorghum, wheat or common bean) and genotype grows for 140 days after emergence in the soil, tillage, weather and management of the project, twice: well-watered and under a drought treatment (a pulse, a terminal, an intermittent or no drought). The model has no random numbers, so the same controls always give the same plant. Light capture and biomass follow Monteith (1977), limited by transpiration as in Tanner and Sinclair (1983); reference evapotranspiration is FAO-56 Penman–Monteith (Allen et al., 1998; Monteith, 1965); soil water uses the van Genuchten curve (van Genuchten, 1980) and root uptake the stress function of Feddes et al. (1978); leaf water potential, turgor and expansion follow a pressure–volume curve and Lockhart (1965), stomata Jarvis (1976), canopy temperature and the crop water stress index Jackson et al. (1981), and the root system grows by rules after Pagès et al. (2014), with carbon allocated by functional equilibrium. The traits that are read out (projected shoot area, convex hull, solidity, senescence, canopy temperature depression, relative water content, root angle, depth and length) are those defined in the two books on phenotyping that the layer starts from (Lorence and Medina-Jimenez, 2022; Panguluri and Kumar, 2013), which define the traits and not the equations. The constants of four species and seven genotypes are typical and editable values of the layer, not data of an experiment. The layer reaches the project: the crop and its library values, the soil, the weather and the toolbar date (which sets the season and a drought outlook for the next days), the mulch, the irrigation and the nitrogen rate. The user can load measured traits, compare, calibrate and export them.

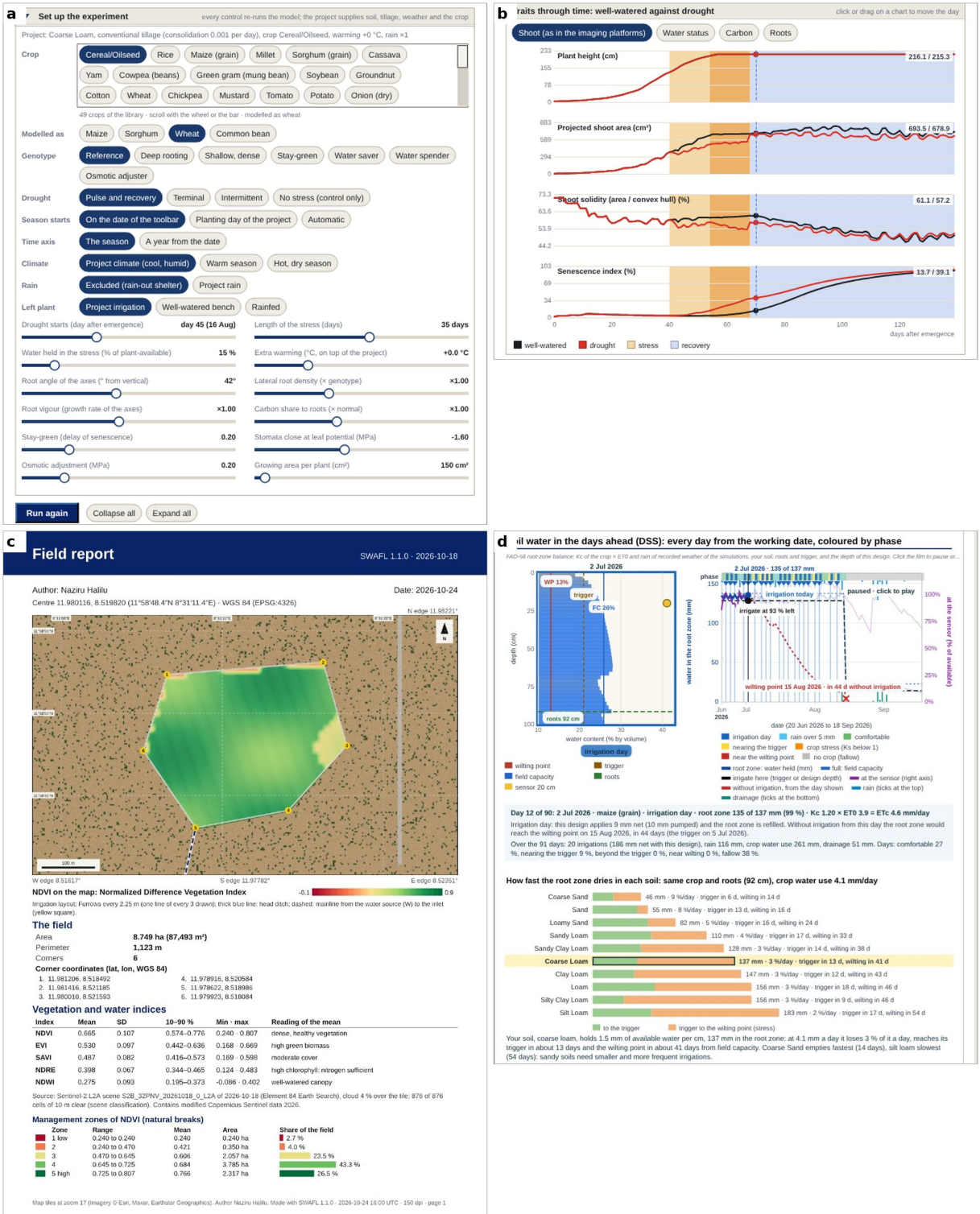


Figure 3: (a) Set-up of the phenotyping experiment (crop, genotype, drought, season, climate, left plant). (b) Traits through time, well-watered against drought, with the phases of the stress marked. (c) The field report (page 1). (d) Soil water in the days ahead with the irrigation decision of the design.

6. What was tested

6.1 Methods

All layers. The application was opened in Chromium (Playwright) and each of the 22 layers was opened, run through the Run button (or the Run simulation button of the design layers) and read: errors of the page script were recorded, the text of the results tab and the number of charts in the results and the charts tabs were counted. The layers with Daisy output were opened in a fresh application to see their empty state.

Crop library. For each of the 110 combinations of crop and density the library gives parameters; the constraints tested were: $K_{mid} \geq K_{ini}$, $0.3 < K_{mid} < 1.4$, positive stages, a season length of 60 to 365 days, $0.1 < p < 0.8$, $0.2 < K_y < 1.6$ where tabulated, a root depth of 20 to 400 cm, a daily K_c curve whose maximum equals the largest of the three coefficients, and no daily jump of more than 0.05.

Phenotyping model. The model was run in the application with a patched entry point that records every run the layer makes (56 runs when the layer opens), and directly for maize in a warm and in a hot, dry season with the seven genotypes, well-watered and under a drought pulse (onset day 40, 28 days at 30 % of the plant-available water, rain-out shelter): 28 runs, each run twice. Determinism is the maximum absolute difference between all stored daily series of the two runs. The soil water balance is $S_{end} - (S_0 + P + I - T - E_s - D)$ with the totals of the run.

6.2 Results

Layers. All 22 layers opened and ran with no error of the page script. The results tab of each layer had a text of 1 346 to 3 419 characters (median 1 670; the phenotyping layer shows its results on the picture and in its own sections) and four to 16 charts; the three design layers have eight in the charts tab, the seasonal layers 16, 10 and 6 (yield reliability, full water balance, nitrogen budget). The Daisy layers showed no figure before a Daisy run, as designed (Paper 5).

Crop library. Of the 110 combinations, 97 satisfy all constraints. The 13 that do not are evergreen tree crops (citrus in six densities, olive in six, date palm in one young planting) for which the library gives a mid-season coefficient below the initial coefficient, which for evergreen crops is the end-of-season value; the youngest olive plantings also have a mid-season coefficient of 0.25

to 0.30, below the general range. These values follow the source (a low ground cover of young trees) and are not errors of the code, but the layer draws a curve that falls from the initial to the mid-season value, which is unusual and is stated here.

Phenotyping. The 56 runs of the layer and the 28 runs of the experiment were identical when repeated (maximum difference 0 in every stored series), and the soil water balance closed with a largest residual of 4×10^{-5} mm. In the warm season the drought pulse reduced the grain mass by 27 to 42 % and in the hot, dry season by 14 to 42 % (Table 2, Fig. 4). The genotype with the largest loss in both seasons is the shallow, dense one (42 %), whose root system reaches 109 to 84 cm instead of 120 cm; the water saver, which closes its stomata early, has the smallest absolute yield but one of the smallest losses (27 and 14 %), and the deep-rooting genotype loses the least in the hot season among those with a full yield (15 %). The osmotic adjuster gives the same grain mass as the reference in this setting: its adjustment changes the water potential and not the dry matter of the pulse. These are the contrasts that the model was built to show, and they follow from its equations; they are not predictions of real genotypes, whose constants are those of the layer.

Table 2: Deterministic phenotyping layer: maize in a warm and in a hot, dry season, seven genotypes, well-watered bench against a drought pulse (onset day 40, 28 days, 30 % of the plant-available water, rain-out shelter). Grain dry mass in model units; the penalty is the loss by the drought.

	Warm season			Hot, dry season		
Genotype	Well-watered	Drought	Loss (%)	Well-watered	Drought	Loss (%)
Reference	515	345	33	331	246	26
Deep rooting	486	329	32	312	264	15
Shallow, dense	516	298	42	330	193	42
Stay-green	542	377	30	353	253	28
Water saver	331	242	27	146	125	14
Water spender	562	336	40	395	288	27
Osmotic adjuster	515	345	33	331	245	26

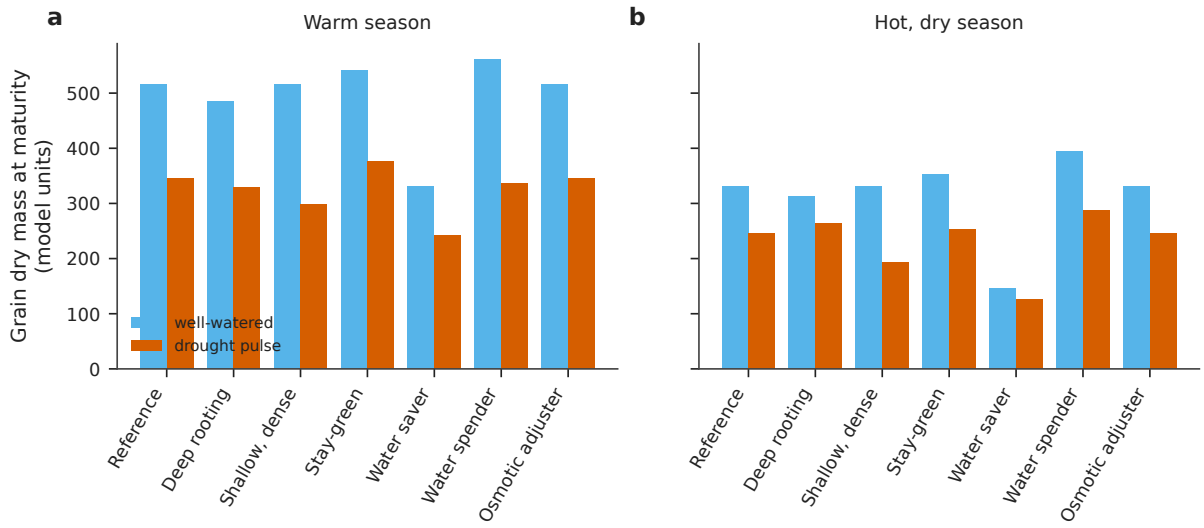


Figure 4: Grain dry mass at maturity of seven genotypes, well-watered and under a drought pulse, in a warm (a) and a hot, dry (b) season.

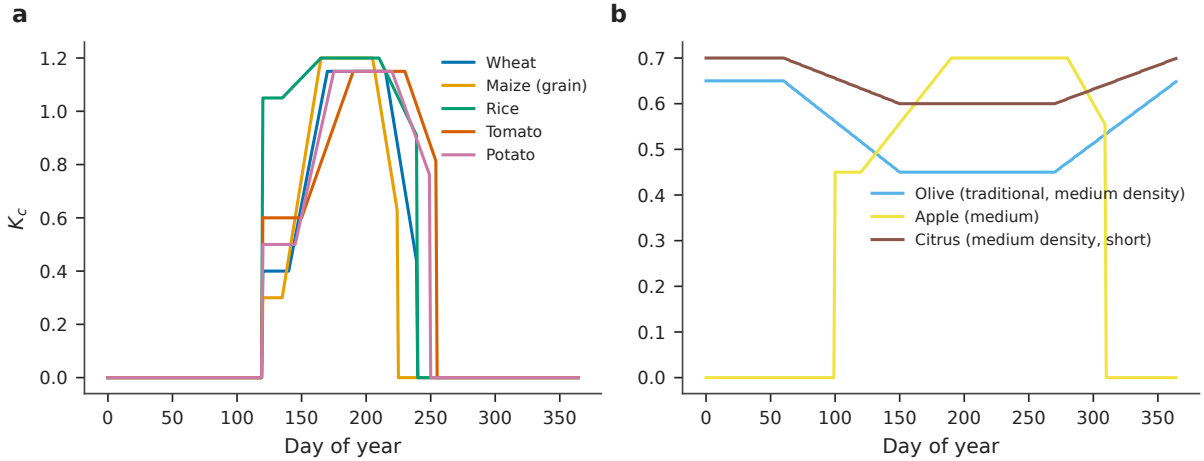


Figure 5: Daily crop coefficient curves of the library (Eq. (2)) at the default sowing date, (a) five annual crops, (b) three perennial crops (medium density).

Daisy. The tests of the Daisy parts are in Papers 4 and 5: the compact designs of the experiments, the soil-column solver, the parsers of the logs, the chemical budget, and the denitrification function, which reproduces 792 stored Daisy outputs after the correction of one defect found by the test.

Reproduction of the earlier papers. After the changes of release 1.1.0, the archived results of Papers 1 and 2 were regenerated and compared with the archived values; Paper 3 is discussed in its own revision (the working date is now today's date, which moved some Engine 2 outputs, not the mass closure).

7. Discussion

What the suite offers. A person can go from a point on the map to a drawn field, to satellite indices and zones, to a design, a schedule, a simulation of the options and a report, in one window and with the same soil, crop and weather throughout, and can follow every number to an equation and its origin.

What the tests do and do not show. They show that the suite starts, runs, is internally consistent in the places tested, and that the parts that were compared with outside references agree. They do not show that the suite is right for a field. In particular: (i) *no field measurement* is used anywhere; the verification is against worked examples, held-out simulations, independent solvers and stored model output (Papers 3 to 5). (ii) *The data services* (Sentinel-2, OpenStreetMap, elevation, SoilGrids, online weather) could not be reached in the environment of the tests; the pictures of the manual and of Fig. 2 were made on generated test images and test answers, so the reading of real scenes and the handling of cloud were not tested here. (iii) The layer sweep checks that each layer runs and produces output, not that every number is right. (iv) The phenotyping model is a model of the forms of the literature with typical constants and has not been fitted or tested against phenotyping data. (v) The crop coefficients are those of FAO-56 and of a review; a local calibration is the user's task. (vi) The test of the interface was made in one browser engine; the Windows application and its installer were not tested here. (vii) The largest class of equations (287 of the labelled) is engineered and has only the tests of this and the earlier papers.

Why traceability. The cost of the rule is the work of writing it down; the benefit is that a disagreement with the suite can be located, and that a reader who disagrees with a choice can change it. Release 1.1.0 found three defects of the first release and one defect of the denitrification function by testing; none of them was visible on the screen.

8. Conclusions

SWAFL Desktop 1.1.0 is a single offline application that joins the field, the satellite, the design, the schedule, the process model and the plant, with traceability of its equations. The release was tested layer by layer, and the tests that could be made (all layers run, crop library, deterministic phenotyping, the Daisy parts) pass or show a stated limit. What remains is the comparison with measurements on real fields, the test of the data services on a connected machine, and the test of

the installer on Windows; each is a defined piece of work and none is claimed.

Software, data and code availability

Software. SWAFL Desktop (version 1.1.0) is open source; the Software source is in the project repository (https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/tree/main/Software) and the program for Windows is SWAFL.exe (https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/releases/latest/download/SWAFL.exe). The whole workbench also runs in a browser, with nothing to install (link in the repository README).

Manual. The Manual (https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/raw/main/Software/v1.1.0/SWAFL_Desktop/docs/manual_v1.1.0/SWAFL_Manual_v1.1.0.pdf) documents every control, screen and equation of the release, with labelled screenshots.

Data. The complete Daisy ensemble outputs (all 10 190 runs, raw and derived) are archived on Zenodo, doi: 10.5281/zenodo.23043626; the run table and a summary of every experiment are in the Data folder (https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/tree/main/Data).

Code. Every check reported here has a script in `Manuscripts/reproducibility/p456` (Node.js scripts that load the shipped code or drive the shipped application, Python scripts for the analysis of the ensemble and the figures); `README_REPRODUCE.md` lists the steps. Daisy itself could not be run in the environment used for these checks, no field measurements are used, and the screenshots are archived rather than regenerated.

Author contributions

N.H. designed the suite, wrote the software and the checks, and wrote the article.

Funding

No funding was received for this work.

Competing interests

The author declares no competing interests.

Acknowledgements

The Daisy model is developed and maintained at the University of Copenhagen. The author thanks the maintainers of the open-source libraries cited.

References

Richard G. Allen, Luis S. Pereira, Dirk Raes, and Martin Smith. Crop evapotranspiration: Guidelines for computing crop water requirements. Technical Report FAO Irrigation and Drainage Paper 56, FAO, Rome, 1998. URL https://www.fao.org/4/x0490e/x0490e00.htm.

Edward M. Barnes, Thomas R. Clarke, Stephen E. Richards, et al. Coincident detection of crop water stress, nitrogen status and canopy density using ground-based multispectral data. In *Proceedings of the 5th International Conference on Precision Agriculture*, Bloomington, MN, 2000.

Jan Doorenbos and Amir H. Kassam. *Yield Response to Water*. FAO Irrigation and Drainage Paper 33, Rome, 1979.

Matthias Drusch, Umberto Del Bello, Sebastien Carlier, et al. Sentinel-2: ESA's optical high-resolution mission for GMES operational services. *Remote Sensing of Environment*, 120:25–36, 2012.

Reinder A. Feddes, Piotr J. Kowalik, and Henryk Zaradny. *Simulation of Field Water Use and Crop Yield*. Pudoc, Wageningen, 1978.

Bo-Cai Gao. NDWI—a normalized difference water index for remote sensing of vegetation liquid water from space. *Remote Sensing of Environment*, 58:257–266, 1996.

Anatoly A. Gitelson, Yoram J. Kaufman, Robert Stark, and Donald Rundquist. Novel algorithms for remote estimation of vegetation fraction. *Remote Sensing of Environment*, 80:76–87, 2002.

Søren Hansen, Per Abrahamsen, Christian T. Petersen, and Merete Styczen. Daisy: model use, calibration, and validation. *Transactions of the ASABE*, 55:1317–1335, 2012. doi: 10.13031/2013.42244.

Alfredo Huete, Kamel Didan, Tomoaki Miura, E. Patricia Rodriguez, Xiang Gao, and Laerte G. Ferreira. Overview of the radiometric and biophysical performance of the MODIS vegetation indices. *Remote Sensing of Environment*, 83:195–213, 2002.

Alfredo R. Huete. A soil-adjusted vegetation index (SAVI). *Remote Sensing of Environment*, 25: 295–309, 1988.

Ray D. Jackson, Sherwood B. Idso, Robert J. Reginato, and Paul J. Pinter. Canopy temperature as a crop water stress indicator. *Water Resources Research*, 17:1133–1138, 1981. doi: 10.1029/WR017i004p01133.

Paul G. Jarvis. The interpretation of the variations in leaf water potential and stomatal conductance found in canopies in the field. *Philosophical Transactions of the Royal Society of London B*, 273:593–610, 1976.

James W. Jones, Gerrit Hoogenboom, Cheryl H. Porter, Kenneth J. Boote, William D. Batchelor, L. A. Hunt, et al. The DSSAT cropping system model. *European Journal of Agronomy*, 18: 235–265, 2003. doi: 10.1016/S1161-0301(02)00107-7.

Brian A. Keating, Peter S. Carberry, Graeme L. Hammer, Merv E. Probert, Michael J. Robertson, Dean Holzworth, et al. An overview of APSIM, a model designed for farming systems simulation. *European Journal of Agronomy*, 18:267–288, 2003. doi: 10.1016/S1161-0301(02)00108-9.

Jack Keller and Ron D. Bliesner. *Sprinkle and Trickle Irrigation*. Van Nostrand Reinhold, New York, 1990.

James A. Lockhart. An analysis of irreversible plant cell elongation. *Journal of Theoretical Biology*, 8:264–275, 1965.

Argelia Lorence and Karla Medina-Jimenez. *Plant Phenotyping: Methods and Protocols*. Humana, New York, 2022.

John L. Monteith. Evaporation and environment. *Symposia of the Society for Experimental Biology*, 19:205–234, 1965.

John L. Monteith. Climate and the efficiency of crop production in Britain. *Philosophical Transactions of the Royal Society of London B*, 281:277–294, 1977.

Loïc Pagès, Grégory Bécsel, Xavier Draye, et al. Calibration and evaluation of ArchiSimple, a simple model of root system architecture. *Ecological Modelling*, 290:76–84, 2014.

Siva K. Panguluri and A. Ashok Kumar. *Phenotyping for Plant Breeding: Applications of Phenotyping Methods for Crop Improvement*. Springer, New York, 2013.

Laura Poggio, Luis M. de Sousa, Niels H. Batjes, Gerard B. M. Heuvelink, Bas Kempen, Eloi Ribeiro, and David Rossiter. SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *SOIL*, 7:217–240, 2021.

John W. Rouse, Robert H. Haas, John A. Schell, and Donald W. Deering. Monitoring vegetation systems in the Great Plains with ERTS. *NASA Special Publication*, 351:309–317, 1974.

Jiří Šimůnek, Martinus Th. van Genuchten, and Miroslav Šejna. Development and applications of the HYDRUS and STANMOD software packages and related codes. *Vadose Zone Journal*, 7:587–600, 2008. doi: 10.2136/vzj2007.0077.

Pasquale Steduto, Theodore C. Hsiao, Dirk Raes, and Elias Fereres. AquaCrop—the FAO crop model to simulate yield response to water: I. concepts and underlying principles. *Agronomy Journal*, 101:426–437, 2009. doi: 10.2134/agronj2008.0139s.

Champ B. Tanner and Thomas R. Sinclair. Efficient water use in crop production: research or re-search? In *Limitations to Efficient Water Use in Crop Production*, pages 1–27. American Society of Agronomy, Madison, WI, 1983.

Martinus Th. van Genuchten. A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Science Society of America Journal*, 44:892–898, 1980. doi: 10.2136/sssaj1980.03615995004400050002x.

W. R. Walker. Guidelines for designing and evaluating surface irrigation systems. Technical Report Irrigation and Drainage Paper 45, Food and Agriculture Organization of the United Nations, Rome, 1989. URL https://www.fao.org/4/t0231e/t0231e00.htm.

David M. Woebbecke, George E. Meyer, Kenneth Von Bargen, and David A. Mortensen. Color indices for weed identification under various soil, residue, and lighting conditions. *Transactions of the ASAE*, 38:259–269, 1995.

A. Checks and scripts

Table 3: Check to script map (scripts in `Manuscripts/reproducibility/p456`).

Check	What is compared	Script → output
All layers	open, run, read	`p6_layer_sweep.js` → `out_p6_sweep.json`
Crop library	constraints and K_c curves	`p6_crop_checks.js` → `out_p6_crops.json`
Phenotyping	determinism, closure, genotypes	`p6_pheno_checks.js` → `out_p6_pheno.json`
Figures and tables		`p456_figs_schematics.py`, `p456_figs_data.py`, `p456_tables.py`